Salma Atef Bakry

Egypt, Cairo
01227000348 salmaabakry06@gmail.com Salma Bakry salma-bakry

PROFESSIONAL SUMMARY

Third-year Computer Science student at Misr International University specializing in Artificial Intelligence, with hands-on experience in full-stack development, embedded systems, and cybersecurity through applied academic projects. Ranked 1st university-wide in the Egyptian Collegiate Programming Contest (ECPC) and currently completing DEPI's Microsoft Machine Learning Engineer track. Active in student leadership through IEEE and ACPC, coordinating technical projects and events.

EDUCATION

Misr International University (MIU) **2024 – 2028**

Bachelor's Degree in Computer Science – ***CGPA: 3.7/4***

PROJECTS

Security Log Analyzer | *C++ | KMP & Boyer-Moore Algorithms*

- Built a cybersecurity log analyzer to detect threats (failed logins, unauthorized access) using predefined patterns.
- Implemented and compared KMP (Dynamic Programming) vs Boyer-Moore (Greedy) for pattern matching performance.

Solar Panel Ordering & Installation Platform | *HTML · CSS · JS · Node.js · Express.js · MongoDB*

- Built a full-stack platform for ordering solar panels and booking installations, supporting Egypt's renewable energy transition.
- Implemented role-based access control (Admin/Customer), appointment scheduling, and personalized solar recommendations.

Ford GoBike Data Analysis & Visualization | *Python · Pandas · PostgreSQL · Plotly · SQLAlchemy*

- Analyzed large-scale bike-sharing trip data to identify patterns in trip duration, usage trends, and rider behavior.
- Designed a PostgreSQL data warehouse using fact and dimension tables and performed SQL-based analytical queries.
- Built interactive visualizations and dashboards using Pandas and Plotly to communicate key insights.

INTERNSHIP / TRAINING

Summer Intern – Commercial International Bank (CIB) **Aug 2026 · 1 Month**

- Selected for the CIB Summer Program under the Gen-AI theme, gaining exposure to banking, digital transformation, AI & ML, data analytics, cybersecurity, sustainable finance, ESG, and professional development.

Microsoft Machine Learning Trainee – DEPI **Jul 2026 · Present**

- Selected for DEPI's Microsoft-sponsored Machine Learning Engineer track.
- Covered advanced Python and data preprocessing/visualization using NumPy, Pandas, Matplotlib, and Seaborn.
- Currently progressing into machine learning concepts and models, deep learning, and computer vision.

Cybersecurity Trainee – National Telecommunications Institute (NTI) **Sep 2025 · 1 Month**

- Covered: Network Fundamentals, Vulnerability Scanning, OWASP Top 10, WordPress Pentesting, Network Penetration Testing, Cloud Security, and SOC Analyst Basics.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, HTML, CSS, JavaScript, SQL
AI/ML & Data: NumPy, Pandas, Matplotlib, Seaborn, Plotly, SQLAlchemy
Web & Databases: Node.js, Express.js, MongoDB, PostgreSQL
Tools & Technologies: Git, GitHub, Linux, Arduino

EXTRACURRICULAR

President – *ACPC Club | MIU* **Present**

- Lead the club's board and coordinate initiatives focused on competitive programming, student development, and ECPC/ACPC preparation.
- Oversee department activities, member engagement, events, and technical initiatives.

Head, R&D Department – *IEEE Student Branch | MIU* **Present**

- Contribute to technical course content and teaching, lead student projects, and support the planning and execution of technical events.

Volunteer Teaching Assistant – *MIU × IEEE Arduino & Programming Workshop* **May 2026**

- Taught Arduino & core programming concepts to school students at a joint MIU–IEEE event.

ECPC Contestant – *Egyptian Collegiate Programming Contest* **2025 & 2026**

- Ranked 1st among MIU participants in ECPC 2026 while competing individually, solving algorithmic problems under time constraints.

CERTIFICATIONS

- Machine Learning Foundations - AWS Academy
- Generative AI - AWS Academy